Panskura Banamali College (Autonomous)

B.Sc.(H) 3rd Semester Examination 2022 (Online mode)

Subject : BCA **Paper: CC5T**

Paper Name : Database Management System

F.M. : 40 **Time : 2 Hours**

Answer any four questions : **4×10=40**

1. Construct an E-R diagram for a COMPANY database with the following requirements :

The company is organized into DEPARTMENTs. Each department has a unique name, unique number and an employee who manage the department. we keep track of the start date of the department manager. The department may have several locations. we store each EMPLOYEE's social security number, address, salary, gender .We also keep track of the direct supervisor of each employee' Each employee may have a number of DEPENDENT's For each dependent, we keep track of their name, gender, DOB and relationship to employee. Give each step in the construction of E-R diagram.

What is an entity super type and subtype, and why is it used? what is subtype discriminator?

7+2+1

2. What is functional dependency? Discuss its types. Explain the role of Functional dependency in the process of normalization. Explain aggregation in detail.

2+3+3+2

3. What is the cardinality of a relation in an E-R diagram? Explain data abstraction with three layers. Why is it important? 3+ 5+2

4. a. What is ternary relation? Explain in detail.
b. what is normalization?

c. Explain strong entity set and weak entity set in detail

d. what is meta data. 4+2+3+1

5. a. Explain functional dependency with example.

b. Explain lossless decomposition.

c. For the given relation R(A,B,C,D) and FD { A->B,A->C,C->D}, R is decomposed in two tables. One is R1(A,B,C) and R2 (C,D) .check whether this decomposition is lossless or lossy.

3+2+5

6. Write short note 5×2

(a) Schema

(b) E-R diagram

(c) normalize the data

(d) DML & DDL

(e) Entities and Attributes

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